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Group 934

Server OS and Service Setup Documentation

This document covers the installation and configuration of the server operating system and core services. Screenshots are inserted under each step to demonstrate the completed configuration and visual proof.

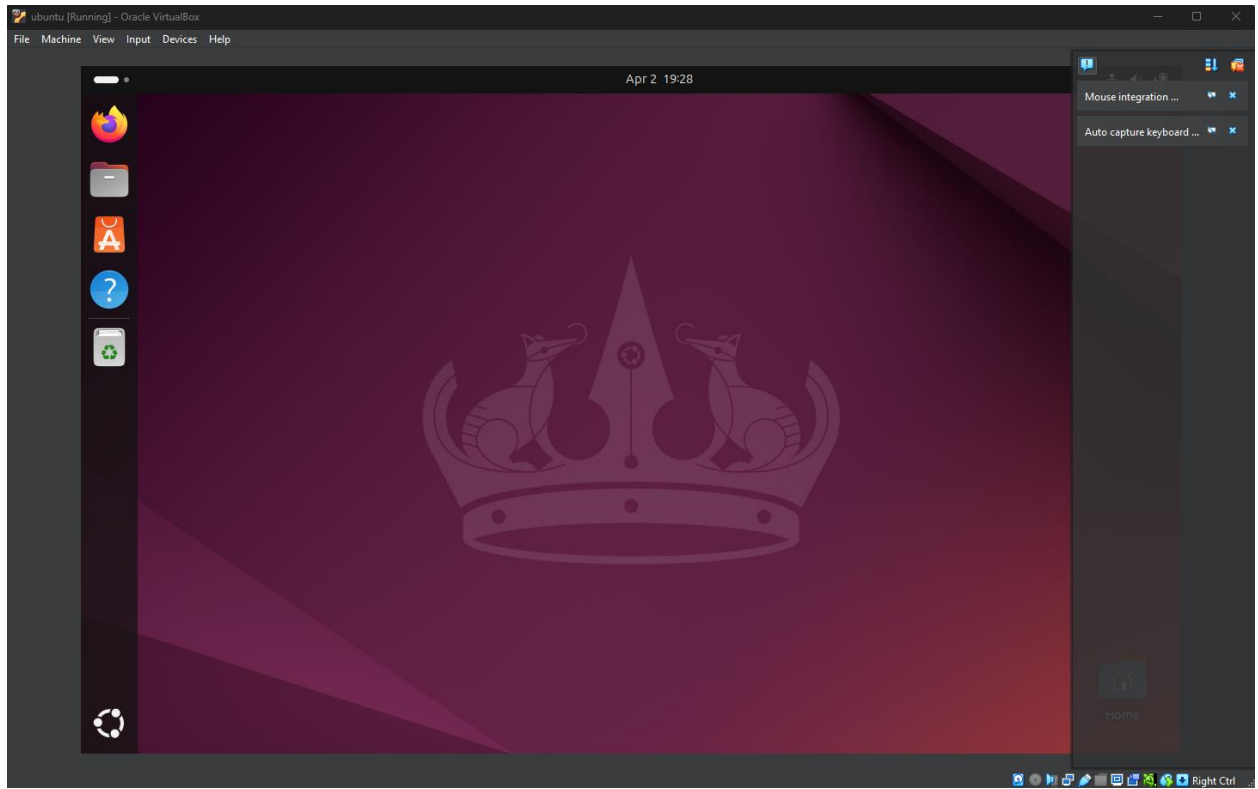
1. Install a Server Operating System

Environment Used:

- Virtualization Software: VirtualBox
- OS: Ubuntu Desktop 24.04 LTS

Steps Taken:

- Downloaded the ISO from ubuntu.com
- Created a new VirtualBox VM
- Allocated CPU, RAM, and storage
- Installed Ubuntu and created the user



2. SSH (Secure Shell)

Installation:

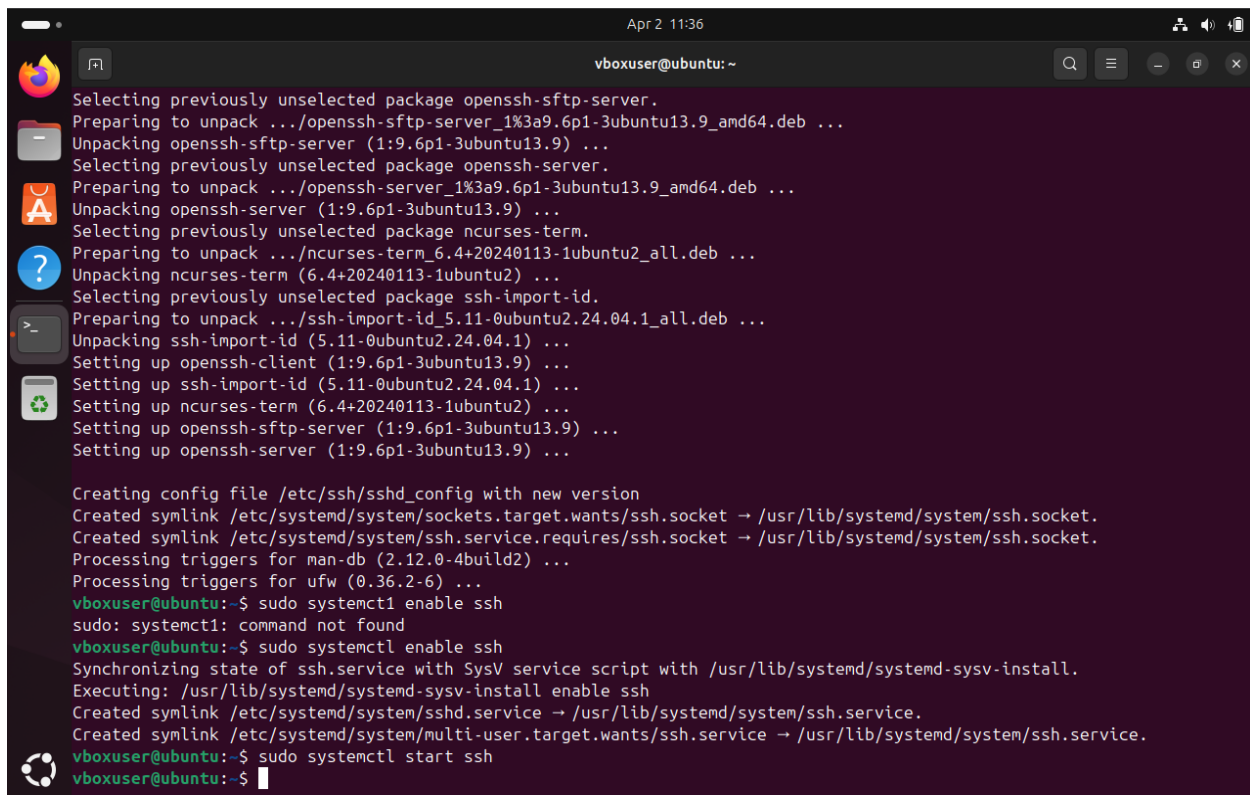
```
sudo apt install openssh-server
```

```
sudo systemctl enable ssh
```

```
sudo systemctl start ssh
```

Testing:

```
ssh vboxuser@localhost
```

A terminal window titled 'vboxuser@ubuntu: ~' showing the installation of SSH. The terminal output includes the following steps: selecting and unpacking openssh-sftp-server, openssh-server, ncurses-term, and ssh-import-id; setting up openssh-client, ssh-import-id, ncurses-term, openssh-sftp-server, and openssh-server; creating a config file /etc/ssh/sshd_config; creating symlinks for /etc/systemd/system/sockets.target.wants/ssh.socket and /etc/systemd/system/ssh.service.requires/ssh.socket; processing triggers for man-db and ufw; attempting to enable ssh with 'sudo systemctl enable ssh' (command not found); synchronizing state of ssh.service with SysV service script; executing 'sudo systemctl enable ssh'; creating symlinks for /etc/systemd/system/sshd.service and /etc/systemd/system/multi-user.target.wants/ssh.service; and finally starting ssh with 'sudo systemctl start ssh'.

```
Apr 2 11:36
vboxuser@ubuntu: ~
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1%3a9.6p1-3ubuntu13.9_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.9) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a9.6p1-3ubuntu13.9_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.9) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-client (1:9.6p1-3ubuntu13.9) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.9) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.9) ...

Creating config file /etc/ssh/sshd_config with new version
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/systemd/system/ssh.socket.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/systemd/system/ssh.socket.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
vboxuser@ubuntu:~$ sudo systemctl enable ssh
sudo: systemctl: command not found
vboxuser@ubuntu:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
vboxuser@ubuntu:~$ sudo systemctl start ssh
vboxuser@ubuntu:~$
```

```
vboxuser@ubuntu:~$ ssh vboxuser@localhost
vboxuser@localhost's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-21-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

74 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

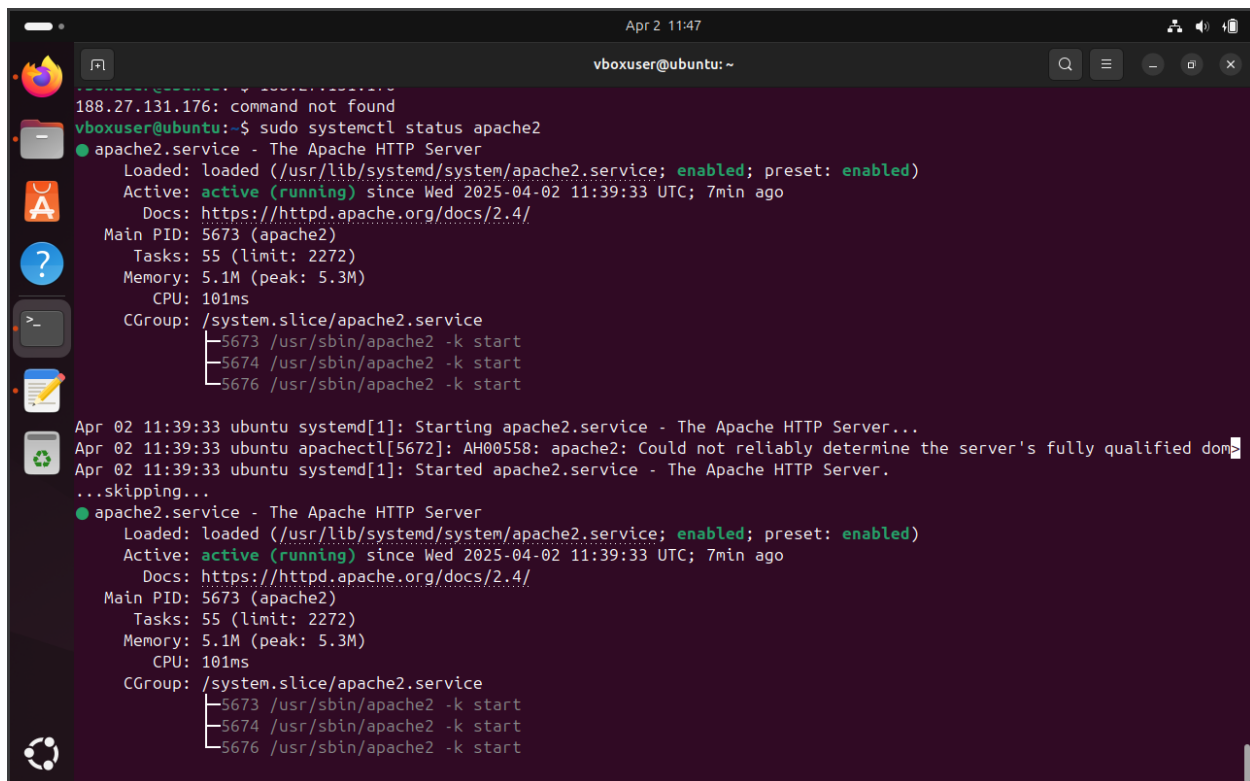
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

3. Web Server (Apache2)

Installation:

`sudo apt install apache2`

Testing: Visit `http://localhost`



A terminal window titled 'vboxuser@ubuntu: ~' showing the installation and status of Apache2. The terminal output includes the command to install Apache2, the status check using 'systemctl', and the service's status details. The status shows that Apache2 is loaded and active (running) since Wed 2025-04-02 11:39:33 UTC. The main PID is 5673, and the tasks are 55 (limit: 2272). The memory usage is 5.1M (peak: 5.3M), and the CPU usage is 101ms. The CGroup is /system.slice/apache2.service, and the tasks are 5673 /usr/sbin/apache2 -k start, 5674 /usr/sbin/apache2 -k start, and 5676 /usr/sbin/apache2 -k start. The terminal also shows the service being started and the status being checked again.

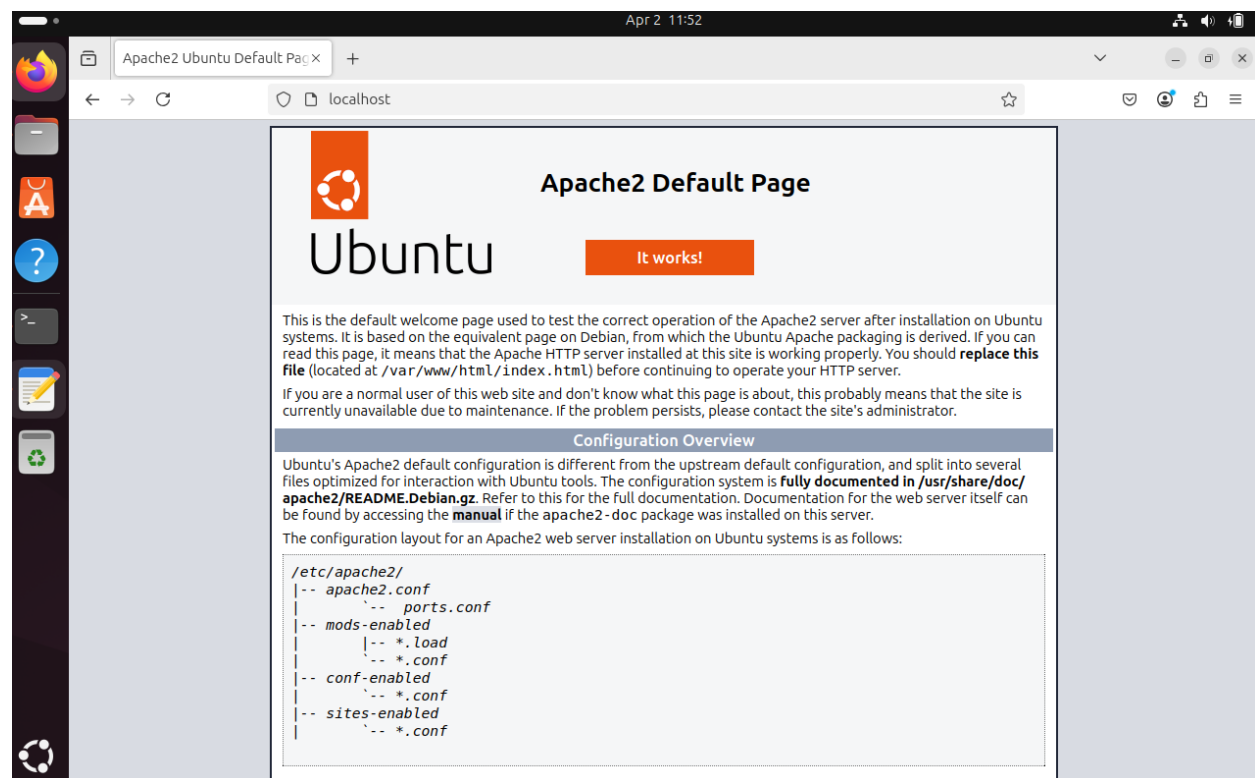
```
188.27.131.176: command not found
vboxuser@ubuntu:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 11:39:33 UTC; 7min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 5673 (apache2)
    Tasks: 55 (limit: 2272)
   Memory: 5.1M (peak: 5.3M)
      CPU: 101ms
   CGroup: /system.slice/apache2.service
           └─5673 /usr/sbin/apache2 -k start
             └─5674 /usr/sbin/apache2 -k start
               └─5676 /usr/sbin/apache2 -k start

Apr 02 11:39:33 ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 02 11:39:33 ubuntu apachectl[5672]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the README file for the appropriate configuration file to edit.
Apr 02 11:39:33 ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
...skipping...
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 11:39:33 UTC; 7min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 5673 (apache2)
    Tasks: 55 (limit: 2272)
   Memory: 5.1M (peak: 5.3M)
      CPU: 101ms
   CGroup: /system.slice/apache2.service
           └─5673 /usr/sbin/apache2 -k start
             └─5674 /usr/sbin/apache2 -k start
               └─5676 /usr/sbin/apache2 -k start
```

```
vboxuser@ubuntu:~$ sudo netstat -tuln | grep 80
tcp6      0      0 :::80      :::*        LISTEN
vboxuser@ubuntu:~$ sudo ufw allow 80
Rules updated
Rules updated (v6)
vboxuser@ubuntu:~$ sudo ufw allow 22
Rules updated
Rules updated (v6)
vboxuser@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
vboxuser@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
80 ALLOW Anywhere
22 ALLOW Anywhere
80 (v6) ALLOW Anywhere (v6)
22 (v6) ALLOW Anywhere (v6)

vboxuser@ubuntu:~$ S
```



4. FTP Server (vsftpd)

Installation:

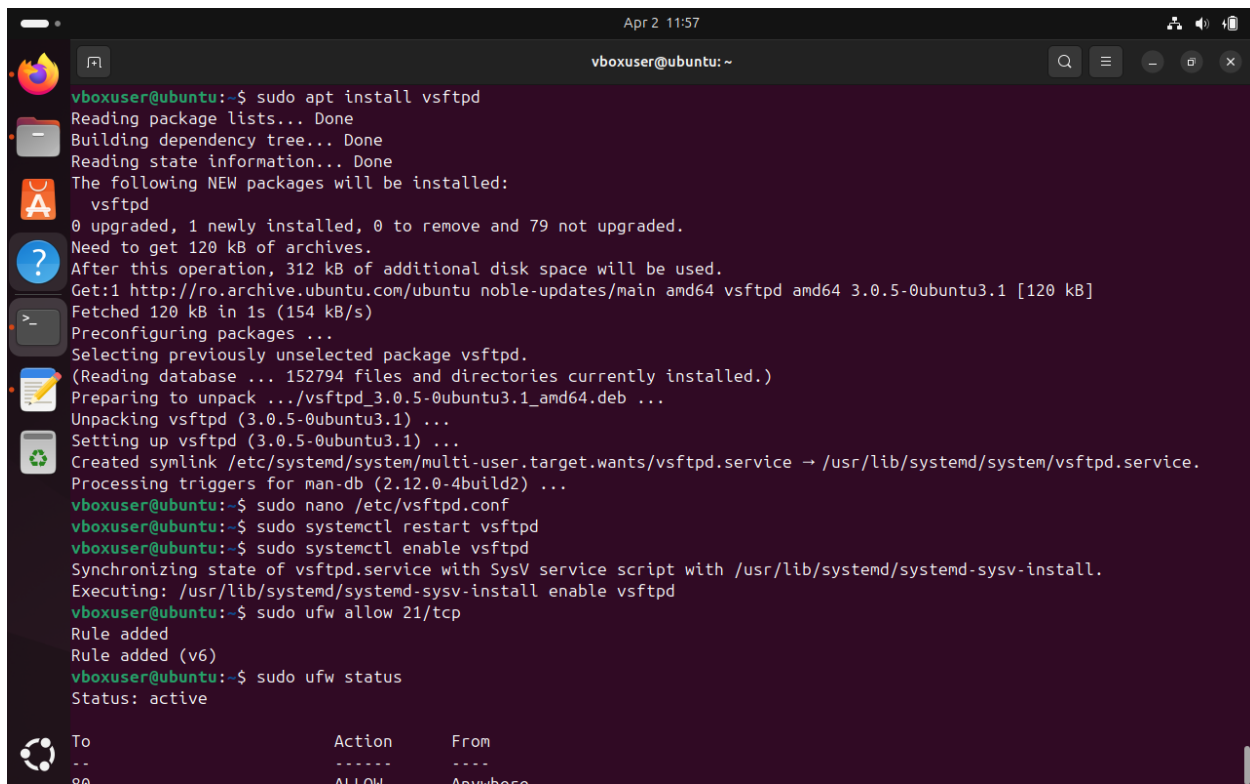
```
sudo apt install vsftpd
```

Enable and Configure:

```
sudo systemctl enable vsftpd
```

```
sudo systemctl start vsftpd
```

Tested using ftp Command



```
vboxuser@ubuntu: ~  
vboxuser@ubuntu:~$ sudo apt install vsftpd  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following NEW packages will be installed:  
  vsftpd  
0 upgraded, 1 newly installed, 0 to remove and 79 not upgraded.  
Need to get 120 kB of archives.  
After this operation, 312 kB of additional disk space will be used.  
Get:1 http://ro.archive.ubuntu.com/ubuntu noble-updates/main amd64 vsftpd amd64 3.0.5-0ubuntu3.1 [120 kB]  
Fetched 120 kB in 1s (154 kB/s)  
Preconfiguring packages ...  
Selecting previously unselected package vsftpd.  
(Reading database ... 152794 files and directories currently installed.)  
Preparing to unpack .../vsftpd_3.0.5-0ubuntu3.1_amd64.deb ...  
Unpacking vsftpd (3.0.5-0ubuntu3.1) ...  
Setting up vsftpd (3.0.5-0ubuntu3.1) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service → /usr/lib/systemd/system/vsftpd.service.  
Processing triggers for man-db (2.12.0-4build2) ...  
vboxuser@ubuntu:~$ sudo nano /etc/vsftpd.conf  
vboxuser@ubuntu:~$ sudo systemctl restart vsftpd  
vboxuser@ubuntu:~$ sudo systemctl enable vsftpd  
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd  
vboxuser@ubuntu:~$ sudo ufw allow 21/tcp  
Rule added  
Rule added (v6)  
vboxuser@ubuntu:~$ sudo ufw status  
Status: active  
  
To Action From  
--  
80 ALLOW Anywhere
```

```
vboxuser@ubuntu:~$ ftp localhost
Connected to localhost.
220 (vsFTPd 3.0.5)
Name (localhost:vboxuser): rob
331 Please specify the password.
Password:
^C
421 Service not available, user interrupt. Connection closed.
ftp: Login failed
ftp>
ftp>
ftp>
ftp>
[1]+  Stopped                  ftp localhost
vboxuser@ubuntu:~$ ftp localhost
Connected to localhost.
220 (vsFTPd 3.0.5)
Name (localhost:vboxuser): vboxuser
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> █
```

5. DHCP Server

Installation:

```
sudo apt install isc-dhcp-server
```

Configuration File:

```
sudo nano /etc/dhcp/dhcpd.conf
```

Modification made:

```
subnet 10.0.2.0 netmask 255.255.255.0 {  
  
    range 10.0.2.50 10.0.2.100;  
  
    option routers 10.0.2.1;  
  
    option domain-name-servers 8.8.8.8;  
  
}
```

```
vboxuser@ubuntu:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 12:17:59 UTC; 1s ago
     Docs: man:dhcpd(8)
    Main PID: 13231 (dhcpd)
      Tasks: 1 (limit: 2272)
    Memory: 3.7M (peak: 4.0M)
       CPU: 19ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─13231 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf enp0s3

Apr 02 12:17:59 ubuntu dhcpd[13231]: Database file: /var/lib/dhcp/dhcpd.leases
Apr 02 12:17:59 ubuntu dhcpd[13231]: PID file: /run/dhcp-server/dhcpd.pid
Apr 02 12:17:59 ubuntu dhcpd[13231]: Wrote 0 leases to leases file.
Apr 02 12:17:59 ubuntu dhcpd[13231]: Listening on LPF/enp0s3/08:00:27:b7:d0:bb/10.0.2.0/24
Apr 02 12:17:59 ubuntu sh[13231]: Listening on LPF/enp0s3/08:00:27:b7:d0:bb/10.0.2.0/24
Apr 02 12:17:59 ubuntu sh[13231]: Sending on LPF/enp0s3/08:00:27:b7:d0:bb/10.0.2.0/24
Apr 02 12:17:59 ubuntu sh[13231]: Sending on Socket/fallback/fallback-net
Apr 02 12:17:59 ubuntu dhcpd[13231]: Sending on LPF/enp0s3/08:00:27:b7:d0:bb/10.0.2.0/24
Apr 02 12:17:59 ubuntu dhcpd[13231]: Sending on Socket/fallback/fallback-net
Apr 02 12:17:59 ubuntu dhcpd[13231]: Server starting service.
```

```
Apr 2 11:59
vboxuser@ubuntu: ~
vboxuser@ubuntu:~$ sudo apt install isc-dhcp-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  isc-dhcp-common
Suggested packages:
  isc-dhcp-server-ldap policycoreutils
The following NEW packages will be installed:
  isc-dhcp-common isc-dhcp-server
0 upgraded, 2 newly installed, 0 to remove and 79 not upgraded.
Need to get 1,281 kB of archives.
After this operation, 4,281 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ro.archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-server amd64 4.4.3-P1-4ubuntu2 [1,236 kB]
Get:2 http://ro.archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-common amd64 4.4.3-P1-4ubuntu2 [45.8 kB]
Fetched 1,281 kB in 1s (1,107 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-server.
(Reading database ... 152850 files and directories currently installed.)
Preparing to unpack .../isc-dhcp-server_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Selecting previously unselected package isc-dhcp-common.
Preparing to unpack .../isc-dhcp-common_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Setting up isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Generating /etc/default/isc-dhcp-server...
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server.service → /usr/lib/systemd/system/isc-dhcp-server.service.
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server6.service → /usr/lib/systemd/system/isc-dhcp-server6.service.
Setting up isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

6. DNS Server (BIND9)

Installation:

```
sudo apt install bind9
```

Testing:

```
nslookup google.com
```

```
Apr 2 12:08
vboxuser@ubuntu: ~
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/system/named.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
vboxuser@ubuntu:~$ sudo systemctl start bind9
vboxuser@ubuntu:~$ sudo systemctl start bind9
vboxuser@ubuntu:~$ sudo systemctl enable bind9
Failed to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
vboxuser@ubuntu:~$ sudo service bind9 start
vboxuser@ubuntu:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 12:07:27 UTC; 34s ago
     Docs: man:named(8)
    Main PID: 13025 (named)
      Status: "running"
       Tasks: 4 (limit: 2272)
      Memory: 7.8M (peak: 8.0M)
         CPU: 71ms
       CGroup: /system.slice/named.service
               └─13025 /usr/sbin/named -f -u bind

Apr 02 12:07:27 ubuntu named[13025]: command channel listening on ::1#953
Apr 02 12:07:27 ubuntu named[13025]: managed-keys-zone: loaded serial 2
Apr 02 12:07:27 ubuntu named[13025]: zone 255.in-addr.arpa/IN: loaded serial 1
Apr 02 12:07:27 ubuntu named[13025]: zone 0.in-addr.arpa/IN: loaded serial 1
Apr 02 12:07:27 ubuntu named[13025]: zone 127.in-addr.arpa/IN: loaded serial 1
Apr 02 12:07:27 ubuntu named[13025]: zone localhost/IN: loaded serial 2
Apr 02 12:07:27 ubuntu named[13025]: all zones loaded
Apr 02 12:07:27 ubuntu named[13025]: running
Apr 02 12:07:37 ubuntu named[13025]: managed-keys-zone: Unable to fetch DNSKEY set '.': timed out
Apr 02 12:07:37 ubuntu named[13025]: resolver priming query complete: timed out
vboxuser@ubuntu:~$

vboxuser@ubuntu:~$ nslookup google.com
Server:
127.0.0.53
Address:
127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.39.46
Name:   google.com
Address: 2a00:1450:400d:80e::200e
vboxuser@ubuntu:~$
```

7. Mail Server (Postfix)

Installation:

`sudo apt install postfix`

Choosed: "Internet Site" during setup

```
vboxuser@ubuntu:~$ sudo apt install postfix
[sudo] password for vboxuser:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libnsl2
Suggested packages:
  mail-reader postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver postfix-mysql postfix-pcre
  postfix-pgsql postfix-sqlite procmail sasl2-bin | dovecot-common
The following NEW packages will be installed:
  libnsl2 postfix
0 upgraded, 2 newly installed, 0 to remove and 79 not upgraded.
Need to get 1,296 kB of archives.
After this operation, 4,321 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ro.archive.ubuntu.com/ubuntu/noble/main amd64 libnsl2 amd64 1.3.0-3build3 [41.4 kB]
Get:2 http://ro.archive.ubuntu.com/ubuntu/noble/main amd64 postfix amd64 3.8.6-1build2 [1,254 kB]
Fetched 1,296 kB in 1s (921 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libnsl2:amd64.
(Reading database ... 153036 files and directories currently installed.)
Preparing to unpack .../libnsl2_1.3.0-3build3_amd64.deb ...
Unpacking libnsl2:amd64 (1.3.0-3build3) ...
Selecting previously unselected package postfix.
Preparing to unpack .../postfix_3.8.6-1build2_amd64.deb ...
Unpacking postfix (3.8.6-1build2)
```

```
Apr 2 17:24
vboxuser@ubuntu: ~
info: Adding group 'postdrop' (GID 128) ...
setting myhostname: ubuntu
setting alias maps
setting alias database
mailname is not a fully qualified domain name. Not changing /etc/mailname.
setting destinations: $myhostname, rob, ubuntu, localhost.localdomain, localhost
setting relayhost:
setting mynetworks: 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
setting mailbox_size_limit: 0
setting recipient_delimiter: +
setting inet_interfaces: all
setting inet_protocols: all
/etc/aliases does not exist, creating it.
WARNING: /etc/aliases exists, but does not have a root alias.

Postfix (main.cf) is now set up with a default configuration. If you need to
make changes, edit /etc/postfix/main.cf (and others) as needed. To view
Postfix configuration values, see postconf(1).

After modifying main.cf, be sure to run 'systemctl reload postfix'.

Running newaliases
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service.
Processing triggers for rsyslog (8.2312.0-3ubuntu9) ...
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
vboxuser@ubuntu:~$ sudo systemctl start postfix
vboxuser@ubuntu:~$ sudo systemctl enable postfix
Synchronizing state of postfix.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable postfix
vboxuser@ubuntu:~$
```

```
vboxuser@ubuntu:~$ sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
   Active: active (exited) since Wed 2025-04-02 17:23:51 UTC; 1min 50s ago
     Docs: man:postfix(1)
    Main PID: 3835 (code=exited, status=0/SUCCESS)
      CPU: 4ms

Apr 02 17:23:51 ubuntu systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Apr 02 17:23:51 ubuntu systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
vboxuser@ubuntu:~$
```

Installation:

Testing:

```
vboxuser@ubuntu:~$ sudo systemctl start dovecot
vboxuser@ubuntu:~$ sudo systemctl enable dovecot
Synchronizing state of dovecot.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable dovecot
vboxuser@ubuntu:~$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 17:25:32 UTC; 58s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
   Main PID: 7877 (dovecot)
    Status: "v2.3.21 (47349e2482) running"
     Tasks: 4 (limit: 2272)
    Memory: 3.2M (peak: 3.4M)
       CPU: 43ms
   CGroup: /system.slice/dovecot.service
           └─7877 /usr/sbin/dovecot -F
             └─7878 dovecot/anvil
               └─7879 dovecot/log
                 └─7880 dovecot/config

Apr 02 17:25:32 ubuntu systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
Apr 02 17:25:32 ubuntu dovecot[7877]: master: Dovecot v2.3.21 (47349e2482) starting up for imap (core dumps disabled)
Apr 02 17:25:32 ubuntu systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email server.
vboxuser@ubuntu:~$
```

```
vboxuser@ubuntu:~$ sudo openssl req -x509 -newkey rsa:4096 -keyout /etc/ssl/private/selfsigned.key -out /etc/ssl/certs/selfsigned.crt -days 365
```

```

vboxuser@ubuntu:~$ sudo nano /etc/dovecot/dovecot.conf
vboxuser@ubuntu:~$ sudo nano /etc/dovecot/conf.d/10-ssl.conf
vboxuser@ubuntu:~$ sudo nano /etc/dovecot/conf.d/10-ssl.conf
vboxuser@ubuntu:~$ sudo systemctl restart dovecot
vboxuser@ubuntu:~$ sudo systemctl restart dovecot
vboxuser@ubuntu:~$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 17:35:31 UTC; 17s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
  Main PID: 8453 (dovecot)
    Status: "v2.3.21 (47349e2482) running"
     Tasks: 4 (limit: 2272)
    Memory: 3.2M (peak: 3.4M)
       CPU: 40ms
    CGroup: /system.slice/dovecot.service
            └─8453 /usr/sbin/dovecot -F
              └─8454 dovecot/anvil
                └─8455 dovecot/log
                  └─8456 dovecot/config

Apr 02 17:35:31 ubuntu systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
Apr 02 17:35:31 ubuntu dovecot[8453]: master: Dovecot v2.3.21 (47349e2482) starting up for imap (core dumps disabled)
Apr 02 17:35:31 ubuntu systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email server.

```

Apr 2 18:02

vboxuser@ubuntu: ~

```

---
read R BLOCK
* OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE LITERAL+ AUTH=PLAIN]
^C
vboxuser@ubuntu:~$ openssl s_client -connect localhost:993
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 C = RO, ST = Cluj, L = Cluj-Napoca, O = funerareluigi, OU = funerareluigi, CN = igi@gmail.com
verify error:num=18:self-signed certificate
verify return:1
depth=0 C = RO, ST = Cluj, L = Cluj-Napoca, O = funerareluigi, OU = funerareluigi, CN = igi@gmail.com
verify return:1
---
Certificate chain
 0 s:C = RO, ST = Cluj, L = Cluj-Napoca, O = funerareluigi, OU = funerareluigi, CN = igi@gmail.com
  i:C = RO, ST = Cluj, L = Cluj-Napoca, O = funerareluigi, OU = funerareluigi, CN = igi@gmail.com
  a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
  v:NotBefore: Apr  2 17:56:44 2025 GMT; NotAfter: Apr  2 17:56:44 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIEFzCCAv+gAwIBAgIUeb6dqSkv1kqQGWTNVs061u0N8QAwDQYJKoZIhvcNAQEL
BQAwZoxCZAJBgNVBAYTALJPMQ0wCwYDVQQIDARDbHVqMRQwEgYDVQQHDAtDbHVq
LU5hcG9jYjYTEWMBQGA1UECgwNZnVuZXJhcmVsdWlnaTEWMBQGA1UECwwNZnVuZXJh
cmVsdWlnaTEOMAwGA1UEAwwFbHVqZ2kxJjAkBgkqhkiG9w0BCQEF2Z1bmVvYXJl
bHVqZ21hZ21haWwY29tMB4XDTE1MDQwMjE3NTY0NFoXDTE2MDQwMjE3NTY0NFow
ZoxCZAJBgNVBAYTALJPMQ0wCwYDVQQIDARDbHVqMRQwEgYDVQQHDAtDbHVqLU5h
cG9jYjYTEWMBQGA1UECgwNZnVuZXJhcmVsdWlnaTEWMBQGA1UECwwNZnVuZXJhcmV
sdWlnaTEOMAwGA1UEAwwFbHVqZ2kxJjAkBgkqhkiG9w0BCQEF2Z1bmVvYXJl
bHVq
-----END CERTIFICATE-----

```

9. Proxy (Squid)

Installation:

```
sudo apt install squid
```

Testing:

```
curl -x http://localhost:3128 http://example.com
```

Configure browser to use: <http://localhost:3128>

```
vboxuser@ubuntu:~$ sudo systemctl status squid
● squid.service - Squid Web Proxy Server
   Loaded: loaded (/usr/lib/systemd/system/squid.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 18:29:42 UTC; 1min 43s ago
     Docs: man:squid(8)
  Main PID: 9962 (squid)
    Tasks: 4 (limit: 2272)
   Memory: 18.0M (peak: 18.5M)
      CPU: 201ms
   CGroup: /system.slice/squid.service
           └─9962 /usr/sbin/squid --foreground -sYC
             └─9965 "(squid-1)" --kid squid-1 --foreground -sYC
               └─9966 "(logfile-daemon)" /var/log/squid/access.log
                 └─9967 "(pinger)"

Apr 02 18:29:42 ubuntu squid[9965]: Using Least Load store dir selection
Apr 02 18:29:42 ubuntu squid[9965]: Set Current Directory to /var/spool/squid
Apr 02 18:29:42 ubuntu squid[9965]: Finished loading MIME types and icons.
Apr 02 18:29:42 ubuntu squid[9965]: HTCP Disabled.
Apr 02 18:29:42 ubuntu squid[9965]: Pinger socket opened on FD 14
Apr 02 18:29:42 ubuntu squid[9965]: Squid plugin modules loaded: 0
Apr 02 18:29:42 ubuntu systemd[1]: Started squid.service - Squid Web Proxy Server.
Apr 02 18:29:42 ubuntu squid[9965]: Adaptation support is off.
Apr 02 18:29:42 ubuntu squid[9965]: Accepting HTTP Socket connections at conn3 local=[::]:3128 remote=[::] FD 12 flags=9
                                listening port: 3128
Apr 02 18:29:43 ubuntu squid[9965]: storeLateRelease: released 0 objects
vboxuser@ubuntu:~$ curl -x localhost:3128 http://example.com
<!doctype html>
<html>
<head>
  <title>Example Domain</title>
```

10. Docker

Installation:

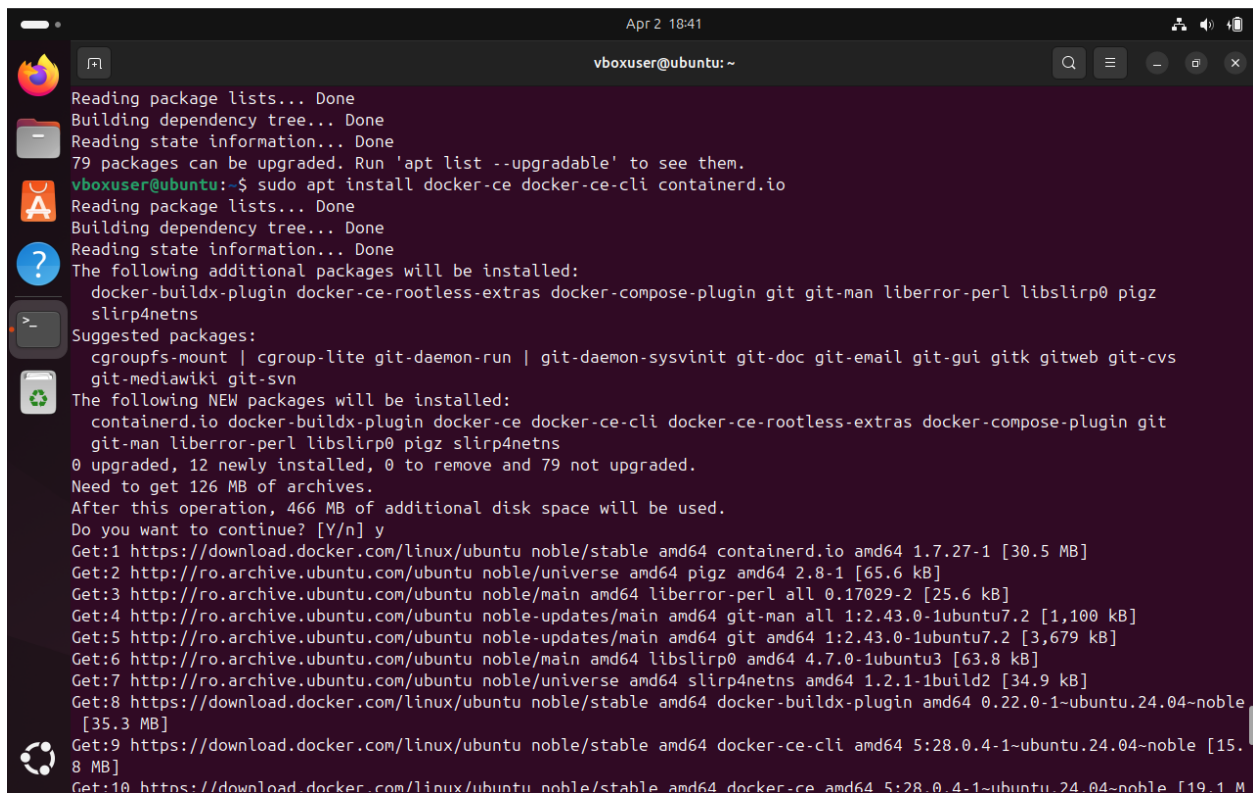
```
sudo apt install docker.io
```

```
sudo systemctl start docker
```

```
sudo systemctl enable docker
```

Test Docker:

```
sudo docker run hello-world
```



```
vboxuser@ubuntu: ~  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
79 packages can be upgraded. Run 'apt list --upgradable' to see them.  
vboxuser@ubuntu:~$ sudo apt install docker-ce docker-ce-cli containerd.io  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  docker-buildx-plugin docker-ce-rootless-extras docker-compose-plugin git git-man liberror-perl libslirp0 pigz  
  slirp4netns  
Suggested packages:  
  cgroupfs-mount | cgroup-lite git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs  
  git-mediawiki git-svn  
The following NEW packages will be installed:  
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin git  
  git-man liberror-perl libslirp0 pigz slirp4netns  
0 upgraded, 12 newly installed, 0 to remove and 79 not upgraded.  
Need to get 126 MB of archives.  
After this operation, 466 MB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 1.7.27-1 [30.5 MB]  
Get:2 http://ro.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]  
Get:3 http://ro.archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl all 0.17029-2 [25.6 kB]  
Get:4 http://ro.archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1:2.43.0-1ubuntu7.2 [1,100 kB]  
Get:5 http://ro.archive.ubuntu.com/ubuntu noble-updates/main amd64 git amd64 1:2.43.0-1ubuntu7.2 [3,679 kB]  
Get:6 http://ro.archive.ubuntu.com/ubuntu noble/main amd64 libslirp0 amd64 4.7.0-1ubuntu3 [63.8 kB]  
Get:7 http://ro.archive.ubuntu.com/ubuntu noble/universe amd64 slirp4netns amd64 1.2.1-1build2 [34.9 kB]  
Get:8 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.22.0-1-ubuntu.24.04-noble  
  [35.3 MB]  
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:28.0.4-1-ubuntu.24.04-noble [15.  
  8 MB]  
Get:10 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:28.0.4-1-ubuntu.24.04-noble [19.1 M
```

```
Apr 2 18:41
vboxuser@ubuntu: ~
Setting up docker-ce (5:28.0.4-1~ubuntu.24.04~noble) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Setting up git (1:2.43.0-1ubuntu7.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
vboxuser@ubuntu:~$ sudo systemctl status docekr
Unit docekr.service could not be found.
vboxuser@ubuntu:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 18:40:33 UTC; 19s ago
     TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
      Main PID: 11716 (dockerd)
        Tasks: 8
       Memory: 29.3M (peak: 29.5M)
          CPU: 757ms
      CGroup: /system.slice/docker.service
              └─11716 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Apr 02 18:40:31 ubuntu dockerd[11716]: time="2025-04-02T18:40:31.785577126Z" level=info msg="detected 127.0.0.53 nameserver"
Apr 02 18:40:31 ubuntu dockerd[11716]: time="2025-04-02T18:40:31.921399581Z" level=info msg="Creating a containerd client"
Apr 02 18:40:32 ubuntu dockerd[11716]: time="2025-04-02T18:40:32.153701016Z" level=info msg="Loading containers: start."
Apr 02 18:40:32 ubuntu dockerd[11716]: time="2025-04-02T18:40:32.986017938Z" level=info msg="Loading containers: done."
Apr 02 18:40:33 ubuntu dockerd[11716]: time="2025-04-02T18:40:33.122997627Z" level=info msg="Docker daemon" commit=6430
Apr 02 18:40:33 ubuntu dockerd[11716]: time="2025-04-02T18:40:33.123323789Z" level=info msg="Initializing buildkit"
Apr 02 18:40:33 ubuntu dockerd[11716]: time="2025-04-02T18:40:33.203933602Z" level=info msg="Completed buildkit initialization"
Apr 02 18:40:33 ubuntu dockerd[11716]: time="2025-04-02T18:40:33.212583262Z" level=info msg="Daemon has completed initialization"
Apr 02 18:40:33 ubuntu dockerd[11716]: time="2025-04-02T18:40:33.212659706Z" level=info msg="API listen on /run/docker.sock"
Apr 02 18:40:33 ubuntu systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```

```
Apr 2 18:42
vboxuser@ubuntu: ~
vboxuser@ubuntu:~$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
vboxuser@ubuntu:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
e6590344b1a5: Pull complete
Digest: sha256:7e1a4e2d11e2ac7a8c3f768d4166c2defeb09d2a750b010412b6ea13de1efb19
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
vboxuser@ubuntu:~$
```

```
Apr 2 18:43
vboxuser@ubuntu: ~

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

vboxuser@ubuntu:~$ sudo docker run -d -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
6e909acdb790: Pull complete
5eaa34f5b9c2: Pull complete
417c4bccf534: Pull complete
e7e0ca015e55: Pull complete
373fe654e984: Pull complete
97f5c0f51d43: Pull complete
c22eb46e871a: Pull complete
Digest: sha256:124b44bfc9ccd1f3cedf4b592d4d1e8bddb78b51ec2ed5056c52d3692baebc19
Status: Downloaded newer image for nginx:latest
0a328264ec130222a3a21f3e6ead5012a3bdd580e18013d0c91f53fb6a605bc5
vboxuser@ubuntu:~$
```

System Administration & Security Tasks Documentation

This document summarizes all administration and security tasks performed on the Ubuntu server, including commands used and brief descriptions. Screenshots will be inserted after each step to provide visual proof.

1. Monitor Services Availability (Up/Down)

Tool: systemctl, monit

Commands Used:

sudo systemctl status apache2

```
vboxuser@ubuntu:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 17:19:33 UTC; 1h 32min ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 1169 (apache2)
      Tasks: 55 (limit: 2272)
     Memory: 6.2M (peak: 7.7M)
        CPU: 694ms
    CGroup: /system.slice/apache2.service
            └─1169 /usr/sbin/apache2 -k start
              └─1172 /usr/sbin/apache2 -k start
                └─1173 /usr/sbin/apache2 -k start

Apr 02 17:19:33 ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 02 17:19:33 ubuntu apachectl[1150]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the README file for the appropriate %FQDN% setting
Apr 02 17:19:33 ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
```

sudo systemctl status ssh

```
vboxuser@ubuntu:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 17:19:33 UTC; 1h 32min ago
 TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1095 (sshd)
       Tasks: 1 (limit: 2272)
      Memory: 1.2M (peak: 2.3M)
         CPU: 41ms
    CGroup: /system.slice/ssh.service
            └─1095 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Apr 02 17:19:32 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Apr 02 17:19:32 ubuntu sshd[1095]: Server listening on :: port 22.
Apr 02 17:19:33 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

sudo systemctl status dovecot

```
vboxuser@ubuntu:~$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-02 18:00:34 UTC; 51min ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
    Main PID: 8832 (dovecot)
   Status: "v2.3.21 (47349e2482) running"
       Tasks: 5 (limit: 2272)
      Memory: 3.6M (peak: 5.9M)
         CPU: 79ms
    CGroup: /system.slice/dovecot.service
            └─8832 /usr/sbin/dovecot -F
                └─8833 dovecot/anvil
                    └─8834 dovecot/log
                        └─8835 dovecot/config
                            └─8842 dovecot/stats

Apr 02 18:00:34 ubuntu systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
Apr 02 18:00:34 ubuntu dovecot[8832]: master: Dovecot v2.3.21 (47349e2482) starting up for imap (core dumps disabled)
Apr 02 18:00:34 ubuntu systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email server.
Apr 02 18:01:02 ubuntu dovecot[8834]: imap-login: Disconnected: Connection closed (no auth attempts in 12 secs): user=<>
Apr 02 18:04:05 ubuntu dovecot[8834]: imap-login: Disconnected: Inactivity (no auth attempts in 180 secs): user=<>, rip>
```

Description: Checks whether each service is currently active and running.

Optional: Install and configure monit

sudo apt install monit

```
vboxuser@ubuntu:~$ sudo apt install monit
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  sysvinit-core
The following NEW packages will be installed:
  monit
0 upgraded, 1 newly installed, 0 to remove and 79 not upgraded.
Need to get 388 kB of archives.
After this operation, 1,101 kB of additional disk space will be used.
Get:1 http://ro.archive.ubuntu.com/ubuntu noble/universe amd64 monit amd64 1:5.33.0-2build2 [388 kB]
Fetched 388 kB in 1s (415 kB/s)
Selecting previously unselected package monit.
(Reading database ... 157830 files and directories currently installed.)
Preparing to unpack .../monit_1%3a5.33.0-2build2_amd64.deb ...
Unpacking monit (1:5.33.0-2build2) ...
Setting up monit (1:5.33.0-2build2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/monit.service → /usr/lib/systemd/system/monit.service.
Processing triggers for man-db (2.12.0-4build2) ...
vboxuser@ubuntu:~$
```

`sudo nano /etc/monit/monitrc`

Enable web interface:

set httpd port 2812 and

allow localhost

Example service monitoring configuration:

check process sshd with pidfile /run/sshd.pid

start program = "/etc/init.d/ssh start"

stop program = "/etc/init.d/ssh stop"

Restart monit:

`sudo systemctl restart monit`

```
vboxuser@ubuntu:~$ sudo monit status
```

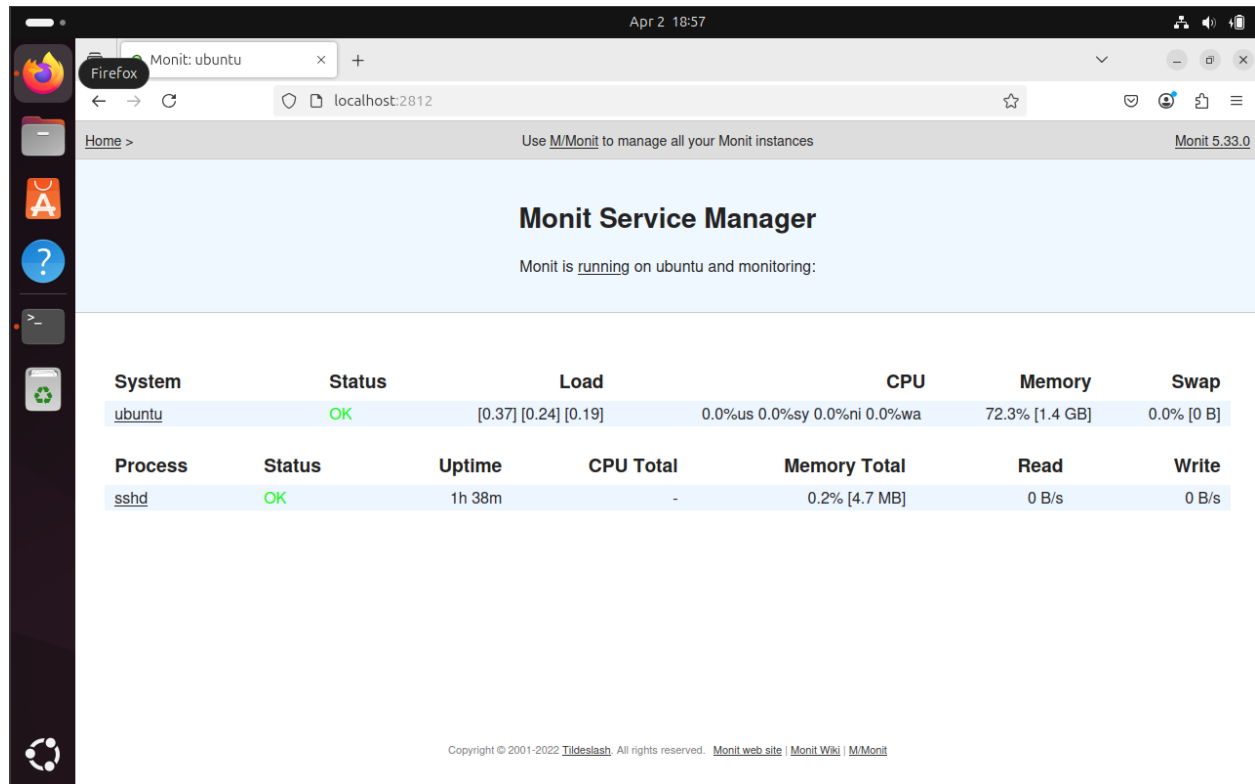
```
Monit 5.33.0 uptime: 0m
```

Process 'sshd'

status	OK
monitoring status	Monitored
monitoring mode	active
on reboot	start
pid	1095
parent pid	1
uid	0
effective uid	0
gid	0
uptime	1h 38m
threads	1
children	0
cpu	-
cpu total	-
memory	0.2% [4.7 MB]
memory total	0.2% [4.7 MB]
security attribute	unconfined
filedescriptors	4 [0.4% of 1024 limit]
total filedescriptors	4
read bytes	0 B/s [71.6 kB total]
disk read bytes	0 B/s [0 B total]
disk read operations	0.0 reads/s [101 reads total]
write bytes	0 B/s [51 B total]
disk write bytes	0 B/s [0 B total]
disk write operations	0.0 writes/s [5 writes total]
data collected	Wed, 02 Apr 2025 18:57:35

System 'ubuntu'

Web interface:



The screenshot shows the Monit Service Manager web interface running in a Firefox browser. The browser's address bar shows the URL `localhost:2812`. The page has a light blue header with the title "Monit Service Manager" and a status message: "Monit is running on ubuntu and monitoring:". Below the header, there are two tables displaying system and process monitoring data.

System Monitoring Table:

System	Status	Load	CPU	Memory	Swap
ubuntu	OK	[0.37] [0.24] [0.19]	0.0%us 0.0%sy 0.0%ni 0.0%wa	72.3% [1.4 GB]	0.0% [0 B]

Process Monitoring Table:

Process	Status	Uptime	CPU Total	Memory Total	Read	Write
sshd	OK	1h 38m	-	0.2% [4.7 MB]	0 B/s	0 B/s

At the bottom of the page, there is a copyright notice: "Copyright © 2001-2022 Tildeslash. All rights reserved. [Monit web site](#) | [Monit Wiki](#) | [M/Monit](#)".

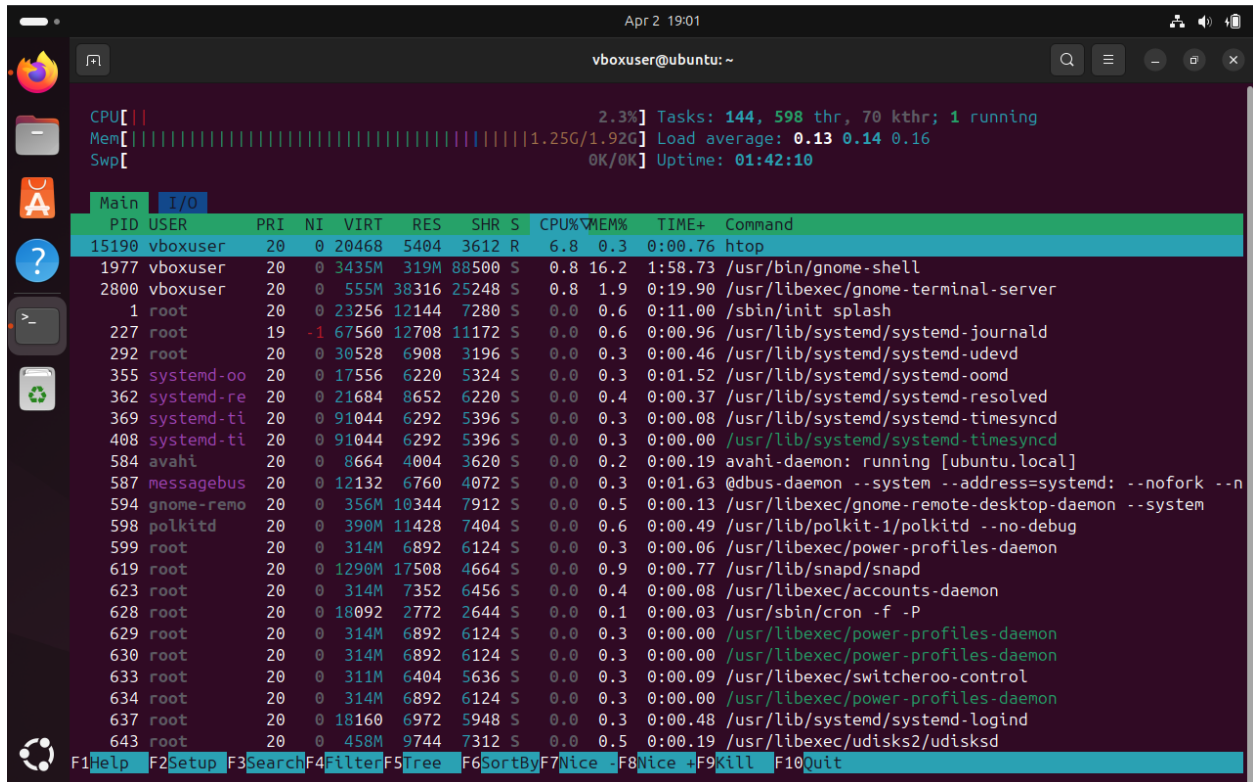
2. Performance Testing

Tools: htop, sar, stress

Commands Used:

sudo apt install htop sysstat stress

htop



The screenshot shows the htop interface in a terminal window. At the top, system statistics are displayed: CPU usage at 2.3%, memory usage at 1.25G/1.92G, and a load average of 0.13, 0.14, and 0.16. The uptime is 01:42:10. Below this, a table lists running processes with columns for PID, USER, PRI, NI, VIRT, RES, SHR, S, CPU%, MEM%, TIME+, and Command. The processes are sorted by CPU usage, with htop (PID 15190) at the top. The terminal window has a dark background with a sidebar on the left containing icons for the application menu, home, search, and other system utilities. The bottom of the window shows a status bar with function key shortcuts like F1 Help, F2 Setup, etc.

```

CPU[ 2.3%] Tasks: 144, 598 thr, 70 kthr; 1 running
Mem[ 1.25G/1.92G] Load average: 0.13 0.14 0.16
Swp[ 0K/0K] Uptime: 01:42:10

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
15190 vboxuser 20 0 20468 5404 3612 R 6.8 0.3 0:00.76 htop
1977 vboxuser 20 0 3435M 319M 88500 S 0.8 16.2 1:58.73 /usr/bin/gnome-shell
2800 vboxuser 20 0 555M 38316 25248 S 0.8 1.9 0:19.90 /usr/libexec/gnome-terminal-server
1 root 20 0 23256 12144 7280 S 0.0 0.6 0:11.00 /sbin/init splash
227 root 19 -1 67560 12708 11172 S 0.0 0.6 0:00.96 /usr/lib/systemd/systemd-journald
292 root 20 0 30528 6908 3196 S 0.0 0.3 0:00.46 /usr/lib/systemd/systemd-udevd
355 systemd-oo 20 0 17556 6220 5324 S 0.0 0.3 0:01.52 /usr/lib/systemd/systemd-oomd
362 systemd-re 20 0 21684 8652 6220 S 0.0 0.4 0:00.37 /usr/lib/systemd/systemd-resolved
369 systemd-ti 20 0 91044 6292 5396 S 0.0 0.3 0:00.08 /usr/lib/systemd/systemd-timesyncd
408 systemd-ti 20 0 91044 6292 5396 S 0.0 0.3 0:00.00 /usr/lib/systemd/systemd-timesyncd
584 avahi 20 0 8664 4004 3620 S 0.0 0.2 0:00.19 avahi-daemon: running [ubuntu.local]
587 messagebus 20 0 12132 6760 4072 S 0.0 0.3 0:01.63 @dbus-daemon --system --address=systemd: --nofork --n
594 gnome-remo 20 0 356M 10344 7912 S 0.0 0.5 0:00.13 /usr/libexec/gnome-remote-desktop-daemon --system
598 polkitd 20 0 390M 11428 7404 S 0.0 0.6 0:00.49 /usr/lib/polkit-1/polkitd --no-debug
599 root 20 0 314M 6892 6124 S 0.0 0.3 0:00.06 /usr/libexec/power-profiles-daemon
619 root 20 0 1290M 17508 4664 S 0.0 0.9 0:00.77 /usr/lib/snapd/snapd
623 root 20 0 314M 7352 6456 S 0.0 0.4 0:00.08 /usr/libexec/accounts-daemon
628 root 20 0 18092 2772 2644 S 0.0 0.1 0:00.03 /usr/sbin/cron -f -P
629 root 20 0 314M 6892 6124 S 0.0 0.3 0:00.00 /usr/libexec/power-profiles-daemon
630 root 20 0 314M 6892 6124 S 0.0 0.3 0:00.00 /usr/libexec/power-profiles-daemon
633 root 20 0 311M 6404 5636 S 0.0 0.3 0:00.09 /usr/libexec/switcheroo-control
634 root 20 0 314M 6892 6124 S 0.0 0.3 0:00.00 /usr/libexec/power-profiles-daemon
637 root 20 0 18160 6972 5948 S 0.0 0.3 0:00.48 /usr/lib/systemd/systemd-logind
643 root 20 0 458M 9744 7312 S 0.0 0.5 0:00.19 /usr/libexec/udisks2/udisksd

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice + F9Kill F10Quit
```

sar -u 1 5

```
vboxuser@ubuntu:~$ sar -u 1 5
Linux 6.11.0-21-generic (ubuntu)      04/02/2025      _x86_64_      (1 CPU)

07:01:43 PM   CPU   %user   %nice   %system   %iowait   %steal   %idle
07:01:44 PM   all    0.00    0.00    2.47    0.00    0.00   97.53
07:01:45 PM   all    0.00    0.00    1.19    0.00    0.00   98.81
07:01:46 PM   all    0.00    0.00    3.53    0.00    0.00   96.47
07:01:47 PM   all    1.27    0.00    3.80    0.00    0.00   94.94
07:01:48 PM   all    1.23    0.00    7.41    0.00    0.00   91.36
Average:      all    0.49    0.00    3.66    0.00    0.00   95.85
```

stress --cpu 2 --timeout 30

```
vboxuser@ubuntu:~$ stress --cpu 2 --timeout 15
stress: info: [15204] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
stress: info: [15204] successful run completed in 15s
```

Description:

- htop: Interactive resource monitor
- sar: Collects CPU usage samples
- stress: Simulates CPU load

3. Log Analyzer

Tools: logwatch, journalctl, less

Commands Used:

sudo apt install logwatch

```
vboxuser@ubuntu:~$ sudo apt install logwatch
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libdate-manip-perl
Suggested packages:
  libsys-cpu-perl libsys-meminfo-perl
The following NEW packages will be installed:
  libdate-manip-perl logwatch
0 upgraded, 2 newly installed, 0 to remove and 79 not upgraded.
Need to get 1,311 kB of archives.
After this operation, 14.2 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ro.archive.ubuntu.com/ubuntu noble/main amd64 libdate-manip-perl all 6.95-1 [923 kB]
Get:2 http://ro.archive.ubuntu.com/ubuntu noble/main amd64 logwatch all 7.7-1ubuntu1 [388 kB]
Fetched 1,311 kB in 0s (2,874 kB/s)
Selecting previously unselected package libdate-manip-perl.
(Reading database ... 157892 files and directories currently installed.)
Preparing to unpack .../libdate-manip-perl_6.95-1_all.deb ...
Unpacking libdate-manip-perl (6.95-1) ...
Selecting previously unselected package logwatch.
Preparing to unpack .../logwatch_7.7-1ubuntu1_all.deb ...
Unpacking logwatch (7.7-1ubuntu1) ...
Setting up libdate-manip-perl (6.95-1) ...
Setting up logwatch (7.7-1ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

sudo logwatch --detail high --mailto example@email.com --
service all --range today

sudo journalctl -xe

```
Apr 02 19:10:11 ubuntu systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
Subject: A start job for unit sysstat-collect.service has begun execution
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit sysstat-collect.service has begun execution.

The job identifier is 6456.
Apr 02 19:10:11 ubuntu systemd[1]: sysstat-collect.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: http://www.ubuntu.com/support

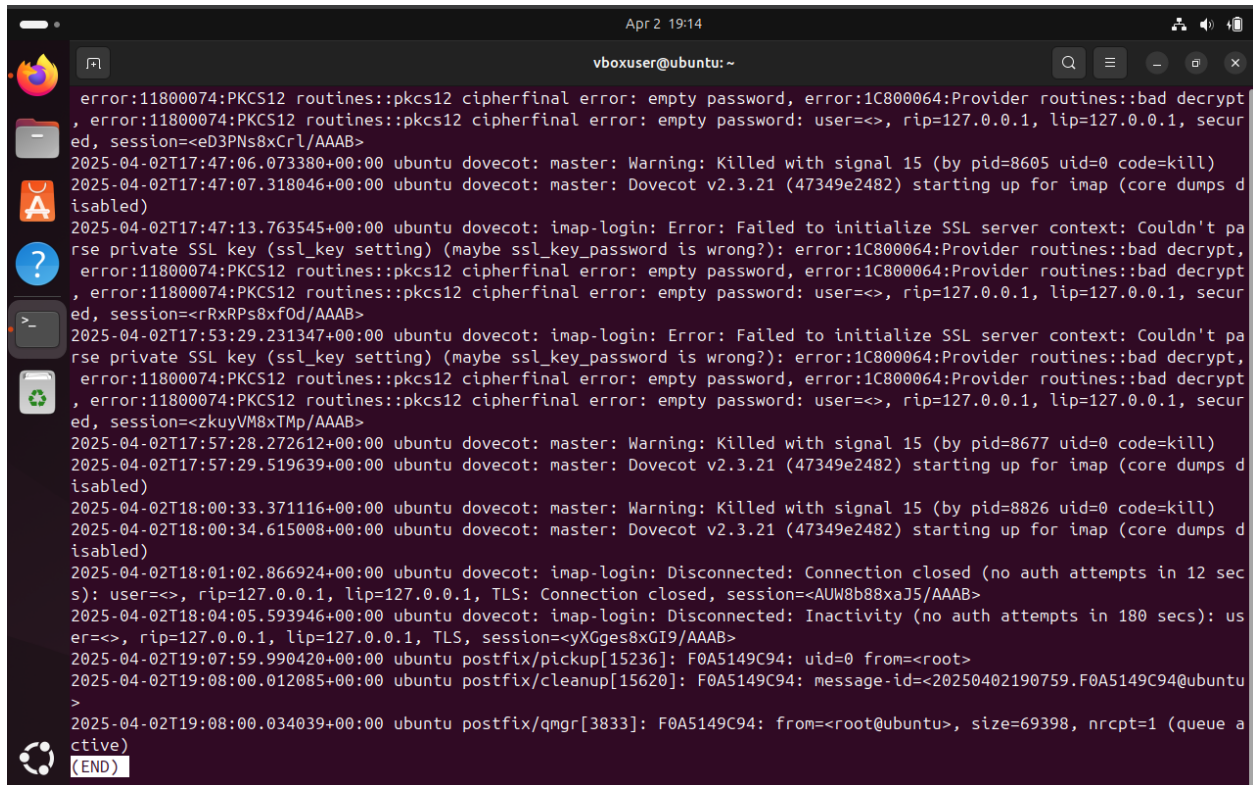
The unit sysstat-collect.service has successfully entered the 'dead' state.
Apr 02 19:10:11 ubuntu systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
Subject: A start job for unit sysstat-collect.service has finished successfully
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit sysstat-collect.service has finished successfully.

The job identifier is 6456.
Apr 02 19:11:45 ubuntu PackageKit[15336]: daemon quit
Apr 02 19:11:45 ubuntu systemd[1]: packagekit.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: http://www.ubuntu.com/support

The unit packagekit.service has successfully entered the 'dead' state.
Apr 02 19:14:48 ubuntu sudo[15693]: vboxuser : TTY=pts/0 ; PWD=/home/vboxuser ; USER=root ; COMMAND=/usr/bin/journalctl
Apr 02 19:14:48 ubuntu sudo[15693]: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
ShowApps -2302/2302 (END)
```

sudo less /var/log/mail.log

A terminal window titled 'vboxuser@ubuntu: ~' with a timestamp of 'Apr 2 19:14'. The window displays the contents of the mail log file, showing various error messages and system events related to Dovecot and Postfix. The log entries include timestamps, hostnames, and detailed error descriptions such as 'cipherfinal error: empty password' and 'Failed to initialize SSL server context'. The terminal window has a dark background and standard Ubuntu window controls on the left and top. The log content is as follows:

```
error:11800074:PKCS12 routines::pkcs12 cipherfinal error: empty password, error:1C800064:Provider routines::bad decrypt
, error:11800074:PKCS12 routines::pkcs12 cipherfinal error: empty password: user=<>, rip=127.0.0.1, lip=127.0.0.1, secur
ed, session=<eD3PNs8xCrl/AAAB>
2025-04-02T17:47:06.073380+00:00 ubuntu dovecot: master: Warning: Killed with signal 15 (by pid=8605 uid=0 code=kill)
2025-04-02T17:47:07.318046+00:00 ubuntu dovecot: master: Dovecot v2.3.21 (47349e2482) starting up for imap (core dumps d
isabled)
2025-04-02T17:47:13.763545+00:00 ubuntu dovecot: imap-login: Error: Failed to initialize SSL server context: Couldn't pa
rse private SSL key (ssl_key setting) (maybe ssl_key_password is wrong?): error:1C800064:Provider routines::bad decrypt
, error:11800074:PKCS12 routines::pkcs12 cipherfinal error: empty password, error:1C800064:Provider routines::bad decrypt
, error:11800074:PKCS12 routines::pkcs12 cipherfinal error: empty password: user=<>, rip=127.0.0.1, lip=127.0.0.1, secur
ed, session=<rRxRPs8xF0d/AAAB>
2025-04-02T17:53:29.231347+00:00 ubuntu dovecot: imap-login: Error: Failed to initialize SSL server context: Couldn't pa
rse private SSL key (ssl_key setting) (maybe ssl_key_password is wrong?): error:1C800064:Provider routines::bad decrypt
, error:11800074:PKCS12 routines::pkcs12 cipherfinal error: empty password, error:1C800064:Provider routines::bad decrypt
, error:11800074:PKCS12 routines::pkcs12 cipherfinal error: empty password: user=<>, rip=127.0.0.1, lip=127.0.0.1, secur
ed, session=<zkuyVM8xTmP/AAAB>
2025-04-02T17:57:28.272612+00:00 ubuntu dovecot: master: Warning: Killed with signal 15 (by pid=8677 uid=0 code=kill)
2025-04-02T17:57:29.519639+00:00 ubuntu dovecot: master: Dovecot v2.3.21 (47349e2482) starting up for imap (core dumps d
isabled)
2025-04-02T18:00:33.371116+00:00 ubuntu dovecot: master: Warning: Killed with signal 15 (by pid=8826 uid=0 code=kill)
2025-04-02T18:00:34.615008+00:00 ubuntu dovecot: master: Dovecot v2.3.21 (47349e2482) starting up for imap (core dumps d
isabled)
2025-04-02T18:01:02.866924+00:00 ubuntu dovecot: imap-login: Disconnected: Connection closed (no auth attempts in 12 sec
s): user=<>, rip=127.0.0.1, lip=127.0.0.1, TLS: Connection closed, session=<AUW8b88xaJ5/AAAB>
2025-04-02T18:04:05.593946+00:00 ubuntu dovecot: imap-login: Disconnected: Inactivity (no auth attempts in 180 secs): us
er=<>, rip=127.0.0.1, lip=127.0.0.1, TLS, session=<yXGges8xGI9/AAAB>
2025-04-02T19:07:59.990420+00:00 ubuntu postfix/pickup[15236]: F0A5149C94: uid=0 from=<root>
2025-04-02T19:08:00.012085+00:00 ubuntu postfix/cleanup[15620]: F0A5149C94: message-id=<20250402190759.F0A5149C94@ubuntu
>
2025-04-02T19:08:00.034039+00:00 ubuntu postfix/qmgr[3833]: F0A5149C94: from=<root@ubuntu>, size=69398, nrcpt=1 (queue a
ctive)
(END)
```

Description:

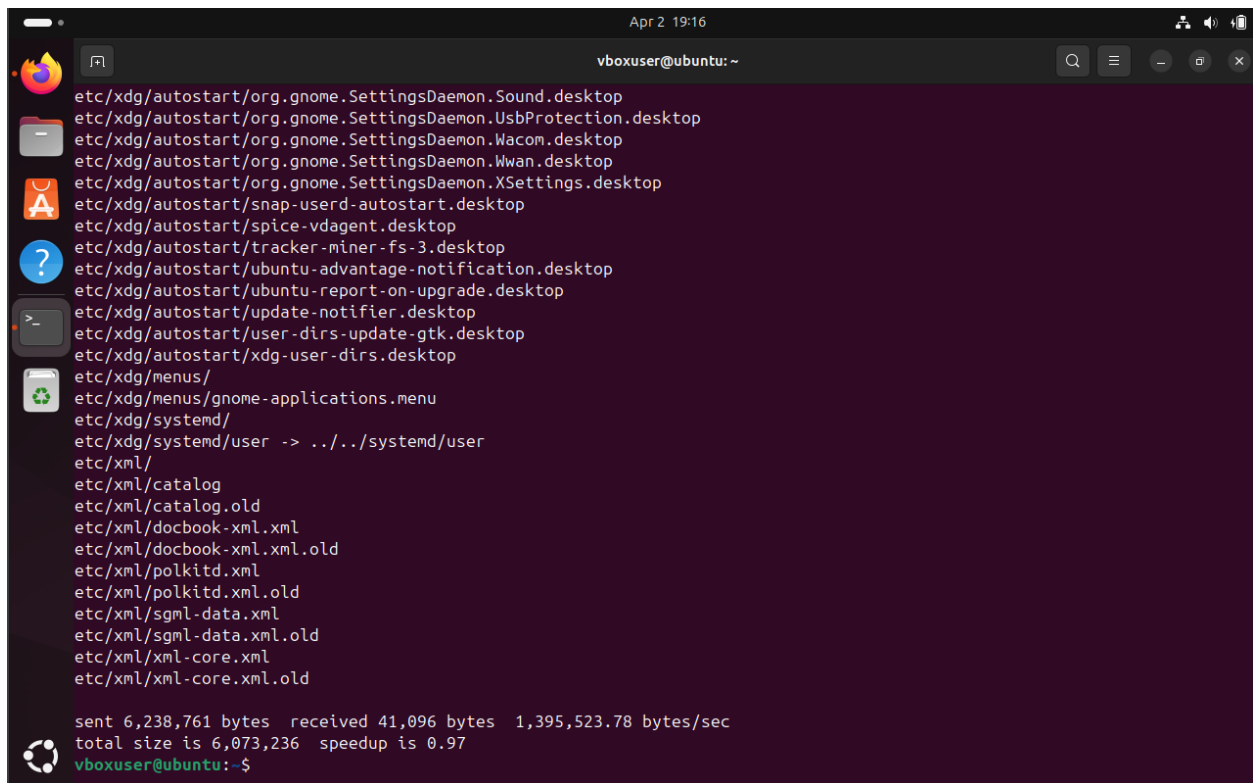
- Analyze system logs
- Useful for diagnosing issues

4. Backup Tasks

Tools: rsync, tar, cron

Commands Used:

`sudo rsync -av /etc /home/user/backups/etc-backup`

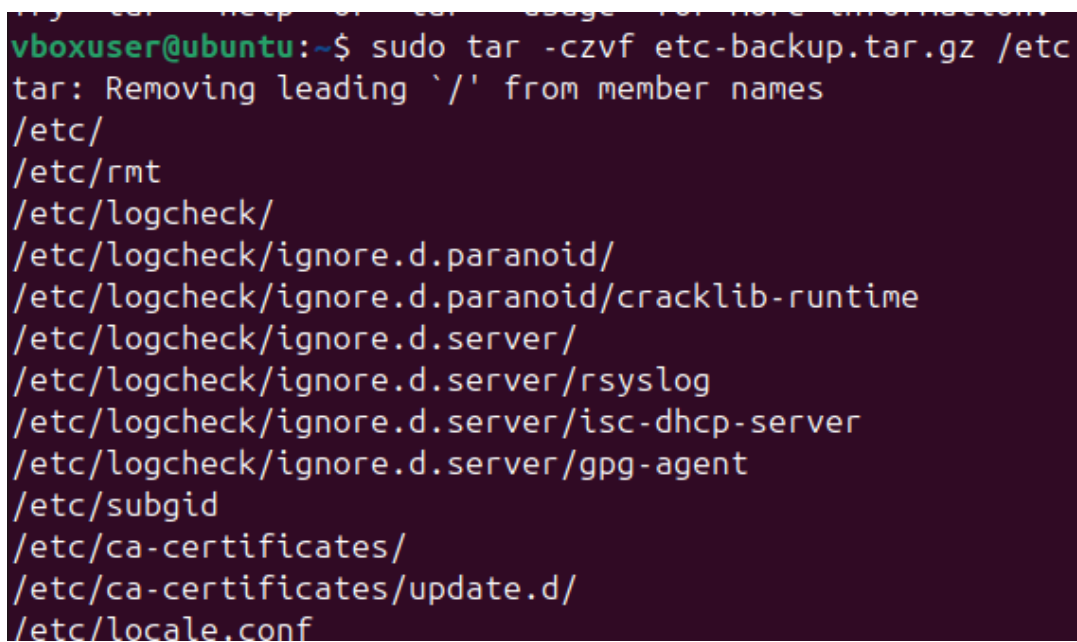


A terminal window titled 'vboxuser@ubuntu: ~' showing the execution of the command `sudo rsync -av /etc /home/user/backups/etc-backup`. The window displays a list of files being synchronized from /etc to the backup directory. The files include desktop files in /etc/xdg/autostart/, menu files in /etc/xdg/menus/, and various XML files in /etc/xml/. The synchronization is complete, showing the total size of the backup and the speedup factor.

```
Apr 2 19:16
vboxuser@ubuntu: ~
etc/xdg/autostart/org.gnome.SettingsDaemon.Sound.desktop
etc/xdg/autostart/org.gnome.SettingsDaemon.UsbProtection.desktop
etc/xdg/autostart/org.gnome.SettingsDaemon.Wacom.desktop
etc/xdg/autostart/org.gnome.SettingsDaemon.Wwan.desktop
etc/xdg/autostart/org.gnome.SettingsDaemon.XSettings.desktop
etc/xdg/autostart/snap-userd-autostart.desktop
etc/xdg/autostart/spice-vdagent.desktop
etc/xdg/autostart/tracker-miner-fs-3.desktop
etc/xdg/autostart/ubuntu-advantage-notification.desktop
etc/xdg/autostart/ubuntu-report-on-upgrade.desktop
etc/xdg/autostart/update-notifier.desktop
etc/xdg/autostart/user-dirs-update-gtk.desktop
etc/xdg/autostart/xdg-user-dirs.desktop
etc/xdg/menus/
etc/xdg/menus/gnome-applications.menu
etc/xdg/systemd/
etc/xdg/systemd/user -> ../../systemd/user
etc/xml/
etc/xml/catalog
etc/xml/catalog.old
etc/xml/docbook-xml.xml
etc/xml/docbook-xml.xml.old
etc/xml/polkitd.xml
etc/xml/polkitd.xml.old
etc/xml/sgml-data.xml
etc/xml/sgml-data.xml.old
etc/xml/xml-core.xml
etc/xml/xml-core.xml.old

sent 6,238,761 bytes  received 41,096 bytes  1,395,523.78 bytes/sec
total size is 6,073,236  speedup is 0.97
vboxuser@ubuntu:~$
```

`sudo tar -czvf etc-backup.tar.gz /etc`



A terminal window with a dark purple background. The prompt is `vboxuser@ubuntu:~$`. The command `sudo tar -czvf etc-backup.tar.gz /etc` has been entered. The output shows the tar command removing the leading slash from member names and listing the files being backed up: `/etc/`, `/etc/rmt`, `/etc/logcheck/`, `/etc/logcheck/ignore.d.paranoid/`, `/etc/logcheck/ignore.d.paranoid/cracklib-runtime`, `/etc/logcheck/ignore.d.server/`, `/etc/logcheck/ignore.d.server/rsyslog`, `/etc/logcheck/ignore.d.server/isc-dhcp-server`, `/etc/logcheck/ignore.d.server/gpg-agent`, `/etc/subgid`, `/etc/ca-certificates/`, `/etc/ca-certificates/update.d/`, and `/etc/locale.conf`.

```
vboxuser@ubuntu:~$ sudo tar -czvf etc-backup.tar.gz /etc
tar: Removing leading '/' from member names
/etc/
/etc/rmt
/etc/logcheck/
/etc/logcheck/ignore.d.paranoid/
/etc/logcheck/ignore.d.paranoid/cracklib-runtime
/etc/logcheck/ignore.d.server/
/etc/logcheck/ignore.d.server/rsyslog
/etc/logcheck/ignore.d.server/isc-dhcp-server
/etc/logcheck/ignore.d.server/gpg-agent
/etc/subgid
/etc/ca-certificates/
/etc/ca-certificates/update.d/
/etc/locale.conf
```

`crontab -e`

Crontab Entry Example:

```
0 2 * * * tar -czf /home/user/backups/etc-$(date +%F).tar.gz
/etc
```

Description: Scheduled backup of /etc using tar. Can be automated with cron.

5. Firewall Configuration

Tool: ufw

Commands Used:

```
sudo apt install ufw
```

```
sudo ufw enable
```

```
sudo ufw allow ssh
```

```
sudo ufw allow 80/tcp
```

```
sudo ufw allow 443/tcp
```

```
sudo ufw status
```

```

vboxuser@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
vboxuser@ubuntu:~$ sudo ufw allow ssh
Rule added
Rule added (v6)
vboxuser@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
vboxuser@ubuntu:~$ sudo ufw allow 443/tcp
Rule added
Rule added (v6)
vboxuser@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
80 ALLOW Anywhere
22 ALLOW Anywhere
21/tcp ALLOW Anywhere
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
443/tcp ALLOW Anywhere
80 (v6) ALLOW Anywhere (v6)
22 (v6) ALLOW Anywhere (v6)
21/tcp (v6) ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)
443/tcp (v6) ALLOW Anywhere (v6)

```

Description: Enables and configures a basic firewall to allow essential services.

6. Brute Force Detection / Prevention

Tool: fail2ban

Commands Used:

sudo apt install fail2ban

sudo systemctl start fail2ban

sudo systemctl enable fail2ban

sudo fail2ban-client status

sudo fail2ban-client status sshd

```
vboxuser@ubuntu:~$ sudo systemctl start fail2ban
vboxuser@ubuntu:~$ sudo systemctl enable fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
vboxuser@ubuntu:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:  sshd
vboxuser@ubuntu:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed:    0
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`- Actions
  |- Currently banned: 0
  |- Total banned:    0
  `-- Banned IP list:
```

Log File:

sudo less /var/log/fail2ban.log

```

2025-04-02 19:19:49,129 fail2ban.server [16457]: INFO -----
2025-04-02 19:19:49,129 fail2ban.server [16457]: INFO Starting Fail2ban v1.0.2
2025-04-02 19:19:49,138 fail2ban.observer [16457]: INFO Observer start...
2025-04-02 19:19:49,150 fail2ban.database [16457]: INFO Connected to fail2ban persistent database '/var/lib/fai
l2ban/fail2ban.sqlite3'
2025-04-02 19:19:49,152 fail2ban.database [16457]: WARNING New database created. Version '4'
2025-04-02 19:19:49,152 fail2ban.jail [16457]: INFO Creating new jail 'sshd'
2025-04-02 19:19:49,460 fail2ban.jail [16457]: INFO Jail 'sshd' uses systemd {}
2025-04-02 19:19:49,462 fail2ban.jail [16457]: INFO Initiated 'systemd' backend
2025-04-02 19:19:49,463 fail2ban.filter [16457]: INFO maxLines: 1
2025-04-02 19:19:49,483 fail2ban.filtersystemd [16457]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.ser
vice + _COMM=sshd'
2025-04-02 19:19:49,483 fail2ban.filter [16457]: INFO maxRetry: 5
2025-04-02 19:19:49,483 fail2ban.filter [16457]: INFO findtime: 600
2025-04-02 19:19:49,483 fail2ban.actions [16457]: INFO banTime: 600
2025-04-02 19:19:49,483 fail2ban.filter [16457]: INFO encoding: UTF-8
2025-04-02 19:19:49,485 fail2ban.jail [16457]: INFO Jail 'sshd' started
2025-04-02 19:19:49,500 fail2ban.filtersystemd [16457]: INFO [sshd] Jail is in operation now (process new journal en
tries)

```

Description: Automatically bans IPs after failed login attempts to prevent brute force attacks.

One such test:

```

vboxuser@ubuntu: ~
vboxuser@ubuntu:~$ ssh fakeuser@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:wHL0HSnbdkVLy7byCIH+JUIGDWEp6abUzQgu50b+09Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:

-----
h v1.0.2
.
l2ban persistent database '/var/lib/fai
ted. Version '4'
l 'sshd'
systemd {}
nd' backend
nal match for: '_SYSTEMD_UNIT=sshd.ser
}
ted
n operation now (process new journal en
tries)
2025-04-02 19:22:44,250 fail2ban.filter [16457]: INFO [sshd] Ignore 127.0.0.1 by ignoreself rule
2025-04-02 19:22:55,410 fail2ban.filter [16457]: INFO [sshd] Ignore 127.0.0.1 by ignoreself rule
~

```